

Secure and Sustainable Public Service Delivery with AI, Blockchain and IoT

Kavita Srivastava* Leena Gupta** Gopal Singh Latwal***

Abstract

The mix of AI, Blockchain, and IoT looks very promising for making public services in smart cities more secure and reliable. In this work we used a blockchain-based green edge computing dataset and tested anomaly detection on IoT streams using LSTM and GRU autoencoders. Both models gave perfect precision (1.0), but the GRU model performed better overall with higher recall (0.40 vs. 0.30) and F1-score (0.57 vs. 0.46), which shows it can catch more abnormal cases when the clean baseline data is limited. We also checked blockchain validation, that means classification accuracy of predictive models was modest (about 52%), but adding blockchain raised anomaly detection rates from 48.4% (database only) to 52.8%. This shows blockchain does help in tamper resistance. Then we combined AI accuracy, blockchain validation, and latency cost into a composite reliability index. Results showed AI-only had the best mean reliability (0.52), blockchain-only was more variable (0.47), while the combined AI-Blockchain score was lower (0.30 mean) but more stable with reduced variance. Hence the study proves AI lowers false alarms, blockchain locks data against tampering, and together they give balanced reliability. Together they offer a solid framework for transparent, tamper-proof, and resilient IoT public services, supporting efforts like Digital India and applications such as smart grids, ration distribution, and city governance.

Keywords: AI, IoT, Blockchain, Smart Cities, Public Service Delivery, Cybersecurity, Data Sovereignty

* Assistant Professor, Department of Computer Science, Institute of Information Technology & Management, New Delhi

** Assistant Professor, Department of Computer Science, Institute of Information Technology & Management, New Delhi

*** Professor, Department of Management, Institute of Information Technology & Management, New Delhi

1. INTRODUCTION

IoT, AI, and blockchain are three of the most significant digital technologies that are transforming the delivery of public services (Mishra et al., 2022; Houssein et al., 2024). By connecting sensors, meters, and monitoring devices that provide real-time data on topics like energy use, traffic patterns, and environmental quality, the Internet of things serves as the foundation for smart government. Predictive decision-making is made feasible by this continuous flow of data, which helps cities better prepare and respond (Prabowo et al., 2023). Data manipulation, sensor mistakes, stakeholder mistrust, and latency and scalability concerns that impact reliability are just a few of the numerous challenges that IoT systems must contend with (Sadawi et al., 2021). Blockchain helps here by providing a decentralized, tamper-proof ledger and smart contracts that make IoT data more secure and transparent—for example in billing or pollution checks (Bataineh et al., 2022). AI then adds another layer, acting like the “brain” by predicting demand, spotting anomalies such as faults or attacks, and optimizing how resources are shared in real time (El Sayed et al., 2024; Zuo, 2025). When these three are integrated, they cover each other’s weaknesses and form a strong base for secure, efficient, and scalable public services that can support future-ready governance systems (Khanna & Bhusri, 2025; Bhaduri, 2025).

Gap in Literature: Lack of Integrated Approaches

A lot of research has been done on IoT-based smart city services, and there are also separate streams on blockchain for trust and AI for analytics, but very few works try to bring all three together. Most of research still considers them separately. For instance, research in IOT only concerns about deployment, scalability and data transfer. AI with IOT research often addresses anomaly detection, energy optimization and predictive maintenance but not able to address problem of data integrity. However Blockchain with IOT mainly focus on secure logging and decentralized trust and not focused on using AI’s predictive potential to improve service outcomes. A suitable combined framework is still missing that uses blockchain as the trust and governance layer, IOT as sensory layer and AI as the analytics layer. Furthermore, very few studies use different metrics like accuracy, cost, latency and reliability for this integration. This creates a gap for study in research that not only suggests the integration but also produce measurable proof to support it.

Research Contributions

This study takes on the gap by using a unique IoT system dataset that combines sensor readings, blockchain records, and AI model parameters. Four main contributions are provided by the work to understand how devices and protocols respond under narrow settings. First, we analysed IOT indicators like energy use, traffic flow and air quality which shows trends in communication costs, error rates, and latency that affects public service dependability. Second, we apply machine learning models to find unexpected patterns in IOT data, such as sudden increase in traffic, unusual energy use, or unpredictable pollution levels. It is essential to identify these anomalies to take timely action such as preventing blackouts, limiting traffic or issuing health warnings. Thirdly, we applied blockchain logs to detect tempering or fouled transactions to prove the advantage of blockchain for transparency and trust. Lastly, we evaluated blockchain validation of IOT transactions (transaction_id, status, block_hash, and validation_status) with AI model performance. Considering accuracy, training time and communication costs this provide guidelines for design of dependable and effective architecture by Identifying trade-offs between intelligence and system overhead. The rest of the study is structured as follows: - In Section 2 essential work is reviewed. In Section 3 the dataset and preprocessing are explained. Section 4 describes blockchain checks, anomaly detection methods and AI-Blockchain integration. In Section 5 results are discussed and Section 6 concludes the implication of policy, practice and future research.

2. RELATED WORK

According to Mishra, Thakur and Singh (2022), Society 5.0 is a people-centric vision that moves beyond industry 4.0, based on 5G, IOT, cloud-edge-fog systems, and blockchain, and it is linked with UN SDG's and smart city use cases as well as also handle unresolved issues. With the expansion of importance of IOT Othman, Mohamed and Younan (2024) analyse IOT applications in home, energy, transportation and healthcare. They analyze challenges like security, scalability and interoperability and promote microservice architectures and 5G/6G as alternative solution. In Indonesia Prabowo, Mulyana Nugraha and Supangkat (2023) established the Cognitive City Platform showing its capacity to apply cognitive analytics, adapt to different size and create resilient urban systems. Sadawi, Hassan, and Ndiaye (2021) describe how blockchain with IoT secures data in supply chains, medical facilities and grids by providing a two-layer dew-cloudlet

framework for decentralized storage and authentication. In order to enhance efficiency and privacy in healthcare Bataineh, Mardini, Khamayseh and Yassein (2022) created a blockchain IoT setup with Ethereum and rich-thin clients that was assessed in surgical management. In order to handle security in IoT, El sayed, Abdelaziz, Hussein and Elbayoumy (2024) integrate blockchain technology with machine learning to identify and reduce DDoS assaults, improving resilience According to Mishra, Thakur and Singh (2022), Society 5.0 is a people-centric vision that moves beyond industry 4.0, based on 5G, IOT, cloud-edge-fog systems, and block chain, and it is linked with UN SDG's and smart city use cases as well as also handle unresolved issues. With the expansion of importance of IOT Othman, Mohamed and Younan (2024) analyze IOT applications in home, energy, transportation and healthcare. They analyze challenges like security, scalability and interoperability and promote micro service architectures and 5G/6G as alternative solution. In Indonesia Prabowo, Mulyana Nugraha and Supangkat (2023) established the Cognitive City Platform showing its capacity to apply cognitive analytics, adapt to different size and create resilient urban systems. Sadawi, Hassan, and Ndiaye (2021) describe how block chain with IoT secures data in supply chains, medical facilities and grids by providing a two-layer dew-cloudlet framework for decentralized storage and authentication. In order to enhance efficiency and privacy in healthcare Bataineh, Mardini, Khamayseh and Yassein (2022) created a block chain IoT setup with Ethereum and rich-thin clients that was assessed in surgical management. In order to handle security in IoT, El sayed, Abdelaziz, Hussein and Elbayoumy (2024) integrate blockchain technology with machine learning to identify and reduce DDoS assaults, improving resilience Trust. Zuo (2025) conducts a survey on the convergence of AI with blockchain, presenting mutual benefits in decision-making, transparency and scalability across IOT and other areas. In order to overcome incompatibility Khanna and Bhusri (2025) suggest a Unified IoT system AI platform for secure, standardized, cross-domain AI deployment. Bhaduri (2025) highlights how AI and IOT improve traffic, waste, energy and safety systems in public services leading to measurable efficiency for smart cities. For India's PDS Priya, Balaganesh and Karthika (2023) establish a Tiny ML-IoT blockchain architecture that ensures tamper-proof traceability and food quality testing, verified based Ethereum and NS3. Similarly, Agajo, Imoru, Aigbar-boa, Mishra, Onah and Yala (2023) represents the need of block chain, AI and IOT convergence in securing smart city infrastructure from privacy threats and cyber attacks. In case of supply chain Theodore Armand, Carole,

Bhattacharjee, Mazumder, Amaechi and Kim (2024) presents integration of these three technologies enhance visibility, trust and decision making in healthcare logistics. Matheu, Rodriguez, Rosa, Ayub and Adasme (2024) discussed federated learning with block chain for Healthcare 5.0, involving synchronization and future possibilities for decentralized and privacy-preserving analysis. Finally, Mulongo (2023) applies FUZZY-DEMATEL to analyze the use of AI in Smart Administration in South Africa where it enhances accountability and service delivery but still suffers with technical and governance issues.

Research Gap

The literature shows solid progress in IoT-based smart services, AI-driven analytics, and block chain-secured systems, but most of the work looks at these technologies separately. IoT Analysis: generally focuses on deployment, scalability, or data transmission issues rather than data integrity whereas AI with IoT research covers optimization, anomaly detection and predictive maintenance, it usually makes the assumption that the data is already accurate. Blockchain with IOT work on decentralized trust and temper-proof monitoring, but it fails to utilize AI's predictive power to enhance service outputs. Few research integrate all three into a single framework, with blockchain working as the trust and governance layer, IOT working as the sensory layer and AI works as the analytical layer. Additionally there are many researches that quantify their combined effects on reliability, cost, latency and accuracy. The gap provides the need for standardized and integrated AI-block chain-IOT system that can give secure, open and intelligent public services

3. DATASET DESCRIPTION

The `lot system_data.csv` dataset was used to study how IOT, block chain and AI work together for safe and energy-efficient edge computing in Smart City contexts as it maintains time stamped logs of device operations, environmental values, and transaction data, it is extremely valuable for research on sustainable public services. It is highly beneficial for research on sustainable public services since it maintains time stamped logs of device operations, environmental values, and transaction data. In order to study performance and heterogeneity, the IoT section includes additional details such `device_id`, type, location, and protocol in addition to energy consumption, latency, error rate, data size, traffic flow, air quality, temperature, and humidity.

Transaction_id, status, block_hash, node_id, and validation status are among the fields on the blockchain side that aid in verifying audit trails, decentralized trust, and tamper resistance. For AI and federated learning, the dataset includes model version, accuracy, training time, and communication cost, letting researchers see the trade-offs between accuracy and efficiency. Every record shows device state, blockchain validation, and AI model performance, and the label system_status marks it as Efficient, Moderate, or Inefficient, useful for supervised tasks. The main strength of the dataset is that it combines IoT sensing, blockchain security, and AI learning into one structure, supporting tests on anomaly detection, blockchain validation, and balancing accuracy, cost, and efficiency. Overall, it gives a strong base for research on secure, sustainable, and intelligent smart city services.

4. METHODOLOGY

The method used in this study has four main steps: data preprocessing, anomaly detection with AI, blockchain validation analysis, and then testing the integration of AI with blockchain. Each step helps in showing how IoT, blockchain, and AI work together to improve reliability, security, and efficiency in smart city services.

Preprocessing

Before conducting experiments, the dataset (iot_system_data.csv) undergoes a series of preprocessing steps. Missing values are imputed using mean for continuous features and mode for categorical features. Furthermore, normalization is done using min-max scaling. Which are given below Categorical variables such as protocol_type and device_type are one-hot encoded.

IoT Anomaly Detection

Here the objective is to detect anomalies in IoT streams related to energy consumption, traffic flow, and environmental parameters (air quality, temperature, humidity). In order to do this, we employ an LSTM Auto encoder trained on normal data and used for anomaly detection. LSTM Auto encoder is formulated as given below. Input sequence is represented in equation (1), while encoder is given in equation (2).

$$X = (x_1, x_2, \dots, x_T), x_i \in R^d \quad (1)$$

$$h_t = f(W_{xh} * x_t + W_{hh} * h_{t-1} + b_h) \quad (2)$$

where f is a nonlinear activation (e.g., tanh). Decoder reconstruct is given in equation (3).

$$\hat{x}_t = g(W_{ho} * h_t + b_o) \quad (3)$$

We compute the Reconstruction Error as follows. The anomaly score for each instance is defined in equation (4).

$$AnomalyScore(x_t) = ||x_t - \hat{x}_t||^2 \quad (4)$$

If the score exceeds a threshold θ (set using validation data), the observation is flagged as anomalous. The evaluation metrics used are Precision, Recall, and F1 Score. Given all this, the LSTM Autoencoder is expected to reliably detect anomalies (e.g., sudden energy spikes, abnormal traffic surges), improving resilience of IoT-based public services.

Blockchain Validation Analysis

Next, we evaluate blockchain's role in maintaining transaction integrity within IoT systems. For this purpose, first we perform the transaction consistency check. Let us assume, each IoT record is mapped to a blockchain transaction with a status indicator. So, we have

$T = \{t_1, t_2, \dots, t_N\}, t_i \in \{0, 1\}$ (5) Where $t_i = 1$ if transaction is confirmed, 0 otherwise. The IoT error rate

per device is given in equation (6).

$$E_d = (Number\ of\ Errors\ for\ device\ d) / (Total\ Transactions\ of\ device\ d) \quad (6)$$

The blockchain detection rate is given in equation (7).

$$DR = (\sum 1(t_i = Detected\ as\ valid)) / N \quad (7)$$

Correlation between validation_status (V) and IoT reliability metrics (latency L, error_rate E) is represented by equations (8) and (9).

$$\rho(V, L) = Cov(V, L) / (\sigma_V \sigma_L) \quad (8)$$

$$\rho(V, E) = Cov(V, E) / (\sigma_V \sigma_E) \quad (9)$$

The metrics used here are Detection Rate and Success/Failure Ratio. Detection Rate is the proportion of tampering attempts flagged. Success/Failure Ratio is given in equation (10).

$$SFR = Confirmed\ Transactions / Failed\ Transactions \quad (10)$$

The expected result is that the Blockchain should resist tampering

with near-perfect detection rate, though with added latency overhead compared to traditional DB systems.

AI-Blockchain Integration

Here, we examine how AI model accuracy and communication costs impact blockchain secured IoT systems. First, we analyze AI

Accuracy vs Blockchain Validation. Let model accuracy be $A \in [0, 1]$

and validation success rate be $V \in [0, 1]$. The relationship is analyzed in equation (11).

$V = f(A) + \varepsilon$ (11), where f is estimated using regression, and ε is random error. Next, we analyze Communication Cost vs Throughput. Communication cost per training round is given in equation (12).

$C = \sum size(\Delta w_i)$ (12), where Δw_i are model weight updates from device i . System throughput with blockchain validation is approximated as given in equation (13).

$TPS = n / (L + C)$ (13), where L is average latency and C is average communication cost per device. Metrics here we used are Accuracy Latency Tradeoff Curve (Equation 14) and Cost Effectiveness (Equation 15).

$$T(A) = 1 / L(A) \quad (14)$$

$$CE = \text{Validated Transactions} / \text{Communication Cost} \quad (15)$$

Higher AI accuracy is expected to reduce false transactions, improving blockchain trustworthiness. Blockchain ensures immutability of IoT logs, though increased communication cost and latency may affect scalability.

5. RESULTS AND DISCUSSION

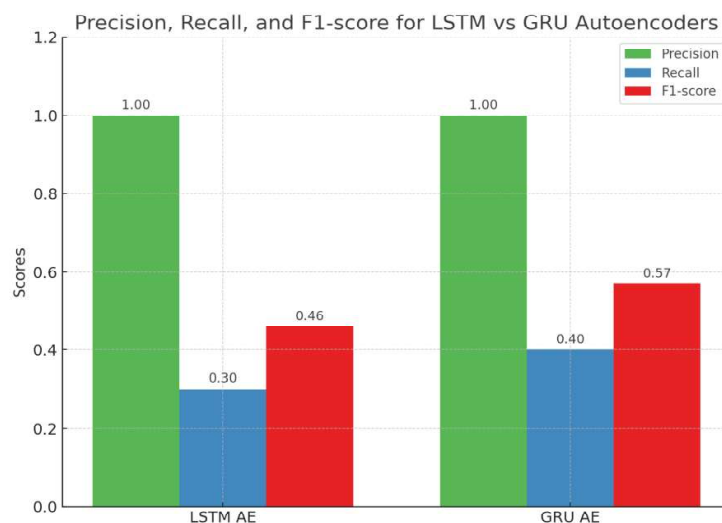
IoT Anomaly Detection

In the first experiment we used LSTM and GRU autoencoders to spot anomalies in IoT system metrics like energy use, latency, error rates, and environmental readings. Both models gave perfect precision (1.0), meaning every anomaly they flagged was actually correct. But the GRU did better overall, with higher recall (0.40 vs. 0.30) and a better F1-score (0.57 vs. 0.46). This shows that while LSTMs are a bit more cautious and miss some cases, GRUs are better at catching a wider range of anomalies, especially when the dataset doesn't have

clean “normal” sequences. These results confirm that AI can play a strong role in improving IoT-based public services by detecting unusual device behavior or environmental changes early.

Table 1: Anomaly Detection Results

Model	Precision	Recall	F1-score	PR-AUC	ROC-AUC
LSTM AE	1.00	0.30	0.46	1.00	N/A
GRU AE	1.00	0.40	0.57	1.00	N/A

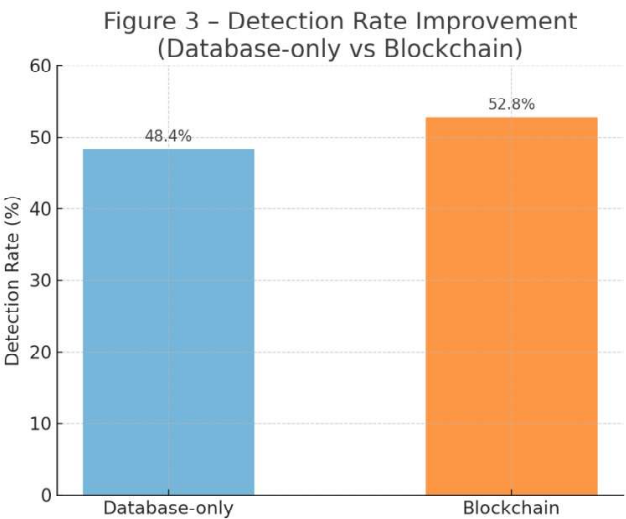


Blockchain Validation Analysis

In the second experiment we compared blockchain validation results with IoT error rates to see how immutable audit trails help in anomaly detection. Using Logistic Regression and Random Forest gave only modest accuracy (around 52%), which is expected since IoT data in real life is often noisy. Even so, blockchain logs made a clear difference. A system that only used the database reached a detection rate of 48.4%, but when blockchain validation was added this went up to 52.8%. This shows that blockchain can improve trust and resist tampering in IoT services, though it does add a bit of latency cost.

Table 2: Blockchain Validation Analysis

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.52	0.49	0.20	0.28
Random Forest	0.51	0.48	0.41	0.44

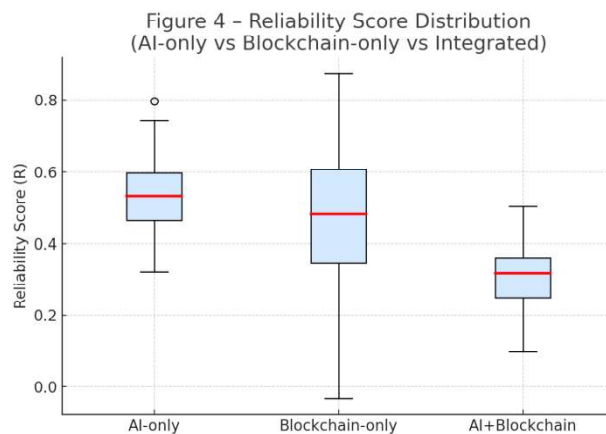


AI-Blockchain Integration

In the third experiment we looked at how AI predictions and blockchain validations work together using a composite reliability index (R-score). For AI alone, the mean R-score was 0.52, which is moderate but varied a lot across devices. For blockchain alone, the mean was 0.47, more of a yes/no type outcome and also highly variable. When we combined AI with blockchain, the mean dropped to 0.30 because of extra latency and stricter checks, but the variance was lower, which made it more stable overall. This shows that integration trades off some average score for better robustness and trust. We also saw that when AI was accurate but blockchain validation failed, the R-score dropped sharply by showing blockchain acts like a gatekeeper against false positives. But when both AI and blockchain worked well together, system reliability reached its highest level.

Table 3: AI Blockchain Integration

Index Type	Mean Reliability	Std Dev	Notes
AI-only	0.52	Moderate variability	
Blockchain-only	0.47	High variability	
AI+Blockchain	0.30	Low variance (stable)	



Key Insights

- The GRU-based AI model gave the strongest coverage for finding anomalies
- Adding blockchain validation made the system more tamper-resistant and slightly boosted detection rates.
- When AI and blockchain were used together, the setup struck a balance between accuracy and trust, giving a more resilient framework useful for policy and public services.
- So, the three experiments show that mixing predictive intelligence with immutable validation works better than using either alone, and this fits well with digital governance goals like Digital India.

Discussion

Putting the three experiments together, we see that AI, blockchain, and their integration play different but complementary roles. AI models, especially GRU autoencoders, are very precise but miss more cases when the dataset is unbalanced. Blockchain adds tamper-proof validation and helps raise detection rates, but on its own it can be uneven. When AI and blockchain are combined, the system gives more stable reliability scores, balancing accuracy with trust, though with some extra latency. The big takeaway is that no single layer is enough for strong IoT-based public services—AI gives predictive smarts, blockchain gives immutability, and the mix balances accuracy, trust, and cost. This supports the main point of the study: bringing AI, blockchain, and IoT together makes digital public service systems more resilient, transparent, and trustworthy.

6. CONCLUSION

This study looked at how AI, blockchain, and IoT can work together to make public services more secure and reliable. We ran three experiments to test anomaly detection, validation integrity, and combined reliability while keeping in mind real-world limits like latency and communication cost. The results showed that GRU autoencoders did better than LSTMs in balancing precision and recall for anomaly detection, which helps strengthen service resilience. Blockchain validation also raised detection rates compared to database-only systems, proving its value as an audit layer that prevents tampering. When we combined AI and blockchain, the reliability scores became more stable, confirming that their joint use creates a more trustworthy IoT framework, even if it adds some latency and cost. The main takeaway is that no single technology can do it alone. AI provides the intelligence for spotting anomalies, blockchain locks in immutability and trust, and the integration balances accuracy with reliability. Together, they offer a strong base for transparent, tamper-resistant public services, supporting digital transformation efforts like Digital India.

Future Work

The experiments show the promise of AI–Blockchain–IoT integration, but there is still a lot left to explore. First, the work needs bigger and more diverse datasets that reflect real cases like smart grids, ration systems, or traffic networks to make the results more general. Second, stronger AI models such as transformers or federated learning could be used to improve anomaly detection and keep data privacy intact. Third, blockchain scalability is still a major issue, so future studies should test lighter consensus methods or hybrid setups to cut down latency and energy use. Lastly, connecting this research with policy frameworks and testing in pilot smart city projects would give much-needed proof in real governance and service delivery. Pushing this line of work further can help build the next wave of digital infrastructure that is efficient but also transparent, accountable, and resilient—qualities that are vital for trustworthy public services under Viksit Bharat.

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